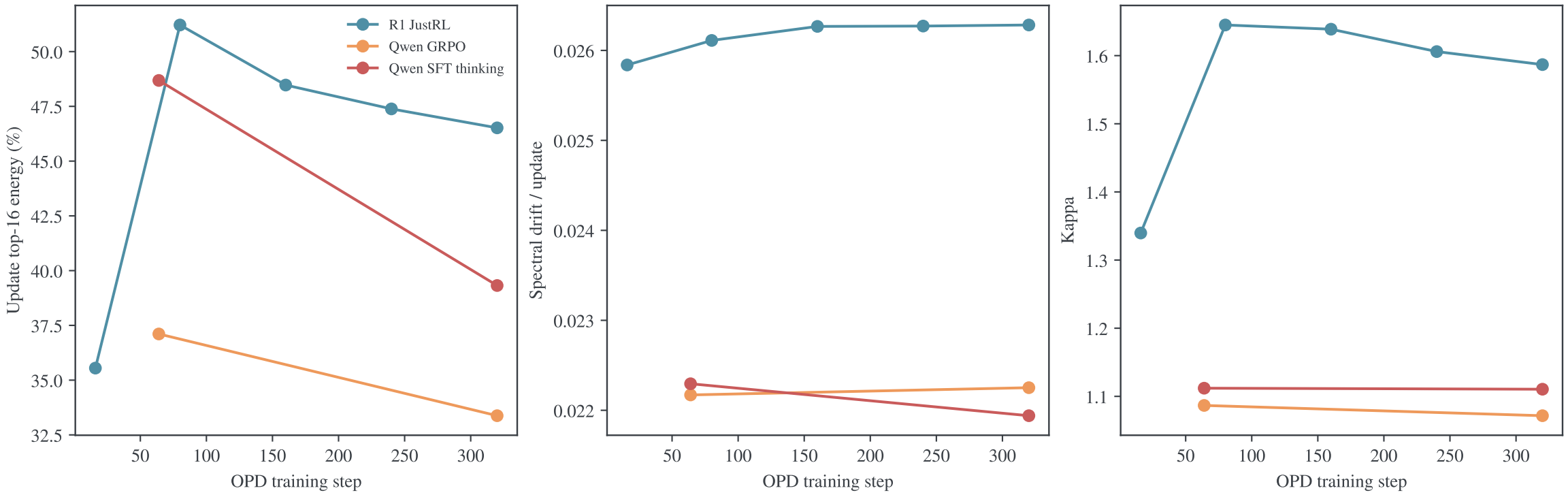
R1 JustRL
Qwen GRPO
Qwen SFT thinking
Update top-16 energy (%)
50.0
47.5
45.0
42.5
40.0
37.5
35.0
32.5
50
100
150
200
250
300
OPD training step
Spectral drift / update
0.026
0.025
0.024
0.023
0.022
50
100
150
200
250
300
OPD training step
Kappa
1.6
1.5
1.4
1.3
1.2
1.1
50
100
150
200
250
300
OPD training step